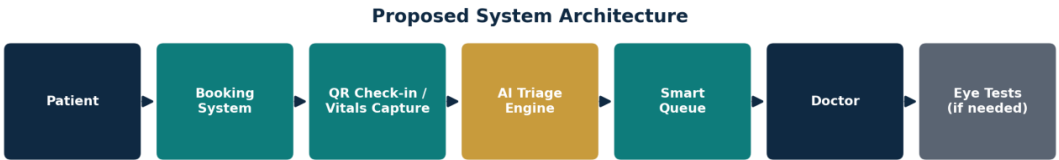


FINAL YEAR PROJECT DOCUMENTATION

AI-Assisted Patient Workflow for Ophthalmic Care

Hashmanis Hospital — Case Study

Reducing waiting times in ophthalmic care through intelligent automation



7 Workflow Steps	3 Triage Classes	23–42 Min., Total Visit	~15–36 Min. Saved / Visit
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1. Executive Summary

Ophthalmic outpatient care spans a wide spectrum of urgency: a patient with sudden vision loss, an eye injury, or a chemical splash needs to be seen within minutes, while a patient attending a routine check-up can safely wait. This document presents a proposed AI-assisted patient workflow for Hashmanis Hospital's ophthalmology service line, designed to reduce total visit time and to route urgent cases ahead of routine ones without removing clinical judgement from the process.

The proposal restructures the patient journey into seven stages — online booking and payment, travel, QR check-in and vitals capture, AI-assisted triage, waiting, doctor consultation, and eye tests where needed. Administrative and sorting tasks that do not require clinical judgement are automated or run in parallel, while every diagnostic and treatment decision remains with the attending doctor. Under this model, a full visit is estimated to take between 23 and 42 minutes depending on whether tests are required, an estimated saving of 15 to 36 minutes compared with a typical manual process.

This is a proposed system design, not a deployed hospital system. All AI features, workflow timings, and savings figures in this document are estimates intended to guide a future pilot, and are clearly marked as such throughout.

2. Introduction

Background of Ophthalmic Patient Care

Ophthalmology outpatient departments handle a broad mix of visit types: emergencies such as sudden vision loss or chemical injury, urgent complaints such as sudden eye pain or redness, and routine visits such as refraction checks, follow-ups, and screening. In most hospital settings, these different visit types enter the same physical queue and are only distinguished once a nurse or doctor manually reviews each patient — a process that depends on staff availability and can introduce delay exactly where delay is most costly.

The Need for Faster Patient Handling

Waiting time in an outpatient eye clinic is shaped less by any single slow step than by the accumulation of small, sequential delays: queueing to pay, queueing to check in, waiting for vitals, and waiting again for a nurse or doctor to assess urgency before the actual consultation begins. Reducing overall visit time therefore requires shortening or parallelizing these administrative steps, not just working faster within any one of them.

The Role of AI-Assisted Workflow

An AI-assisted workflow, as proposed in this document, targets exactly this class of problem. Booking and payment move online, check-in is automated through QR codes, and an AI triage step reads already-collected symptom and vitals data to assign a priority label in under a minute — work that would otherwise require a dedicated manual review queue. Crucially, the system is designed to assist sorting and administration only; diagnosis and treatment decisions remain the doctor's responsibility at every step.

3. Problem Statement

In a conventional, manually-coordinated outpatient process, patient flow is slowed by delays at several distinct points, each of which compounds the others rather than being absorbed by the system:

- Check-in delay — patients queue at a front desk for manual record lookup and data entry before anything else can happen.
- Payment delay — walk-in, counter-based payment adds a further queue before a patient is even registered for the day's visit.
- Triage delay — urgency is assessed only when a nurse or doctor becomes available to review each patient in turn, so sorting happens late and inconsistently rather than as soon as symptom and vitals data exists.
- Administrative overhead — receipts, physical forms, and repeated manual entry of the same patient details consume staff time that could otherwise go toward patient-facing work.
- Undifferentiated waiting — because sorting happens late, an urgent case can end up waiting behind several routine cases purely because of arrival order.

This document proposes a system design intended to address these specific points of delay. It is important to distinguish this proposal clearly from an actual deployed hospital system: nothing described here is currently running at Hashmanis Hospital, and all timings are planning estimates rather than measured operational data.

Note: *This document describes a proposed system design intended for academic and planning purposes. It is not a description of a system currently deployed at Hashmanis Hospital or any other facility.*

4. Objectives

The proposed workflow is designed around five specific, measurable objectives:

- Reduce patient waiting time by removing or parallelizing administrative steps that do not require staff judgement.
- Improve overall patient flow so that urgent cases are identified and routed ahead of routine ones as early as possible in the visit.
- Automate routine administrative steps — booking, payment, and check-in — so front-desk and clinical staff spend less time on repetitive data entry.
- Assist with patient triage using a rules- and pattern-based system that reads symptoms, vitals, and history to suggest a priority label.
- Keep the doctor responsible for every final clinical decision, so automation never substitutes for medical judgement at the point of diagnosis or treatment.

5. Why Hashmanis Hospital?

Eye Care Demands Speed

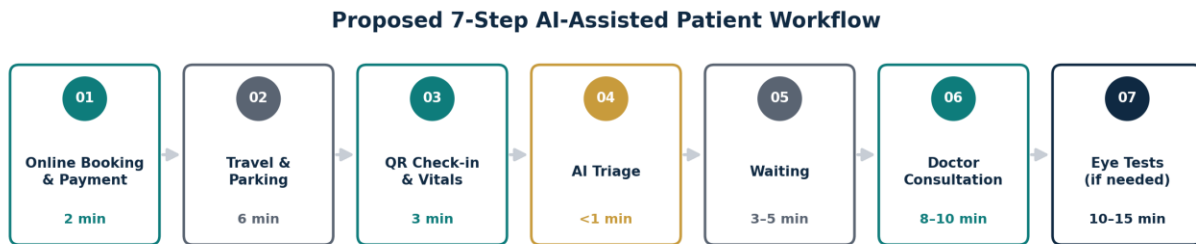
Ophthalmic emergencies — sudden vision loss, penetrating eye injury, or chemical splash to the eye — require attention within minutes, since delay can directly affect the outcome for the patient's vision. Routine visits, such as follow-ups or refraction checks, can comfortably wait without any risk to the patient. This wide spread between urgent and routine cases is precisely the condition under which automated triage is most useful: the system does not need to treat every patient identically, only to reliably flag the minority who cannot wait.

A Suitable Candidate for AI-Assisted Triage

Hashmanis operates as a multi-branch Karachi hospital group, which is relevant for two practical reasons. First, patient volume across branches is high enough that even modest percentage gains in throughput translate into meaningful time savings across many visits. Second, a shared, multi-branch structure means a single validated triage design can be applied consistently across locations, rather than being redesigned separately for each branch.

6. Proposed 7-Step AI Workflow

The proposed patient journey is organized into seven stages. Administrative and sorting steps are digital or automated; only the consultation and, where needed, testing require the doctor's direct time. The figure below summarizes the sequence and estimated duration of each stage.



Total: 23–27 min without tests · 33–42 min with tests (estimated, proposed model)

Figure 6.1 — Proposed 7-step AI-assisted patient workflow (estimated durations).

Step	Stage	Description	Est. Duration
01	Online Booking & Payment	Patient books an ophthalmology slot and pays online before arriving, capturing the stated reason for visit as an input to later triage.	2 min
02	Travel & Parking	Patient-controlled time outside the hospital process, included only so total visit-time estimates reflect a realistic door-to-door figure.	6 min

Step	Stage	Description	Est. Duration
03	QR Check-in & Vitals	Patient scans a QR code to confirm arrival and pull up their record instantly; a technician records vitals into the same digital record.	3 min
04	AI Triage	System reads symptoms, vitals, and history, and assigns a priority label — emergency, urgent, or routine — with a suggested route.	<1 min
05	Waiting	A short buffer for the next doctor to become free; patients are already sorted by priority, so this is not a first-come-first-served queue.	3–5 min
06	Doctor Consultation	Doctor reviews the AI label and underlying data, examines the patient, and independently confirms, adjusts, or overrides the triage.	8–10 min
07	Eye Tests (if needed)	Further diagnostics — imaging, visual field testing, refraction — carried out on the doctor's instruction after the consultation.	10–15 min

Total: 23–27 minutes without tests · 33–42 minutes with tests (estimated)

Note: All AI features are a proposal. All times are estimates based on planning assumptions, not measured clinical data.

7. AI Triage Module

The AI triage module is the single automated decision point in the proposed workflow. It is designed to organize information for the doctor, not to make a clinical decision on its own.

Symptom Input

The reason for visit, captured at the online booking stage, provides the module's first input — for example, sudden vision loss, redness, pain, or a routine follow-up reason.

Vitals and Available Records

Vitals recorded at check-in (such as visual acuity, intraocular pressure, and blood pressure where relevant) and any available history from prior visits are added as further structured inputs.

Emergency / Urgent / Routine Classification

Using these inputs, the module assigns one of three priority labels — emergency, urgent, or routine — based on predefined rules and pattern-matching against known symptom and vitals combinations.

Automatic Routing

The assigned label determines the patient's position in the smart queue, so that emergency and urgent cases are surfaced to the next available doctor ahead of routine cases.

Note: *The AI Triage Module is a proposed decision-support tool. It is explicitly not a replacement for a doctor: it produces a suggested label and routing, both of which are reviewed and can be overridden at the consultation stage.*

8. Human-in-the-Loop / Doctor's Role

The proposed design keeps a human clinician as the final authority at every point where patient care is affected. Specifically, the doctor:

- Examines the patient directly, independent of what the AI module suggested.
- Confirms or overrides the AI triage label — an override takes precedence immediately and is logged.
- Makes the diagnosis, drawing on clinical examination and professional judgement.
- Decides on treatment, including whether further eye tests are required.

This human-in-the-loop structure means the AI module's output functions as a pre-organized starting point for the doctor's attention, never as a constraint on it.

9. Time-Saving Analysis

Estimated savings come from three sources acting together: removing the counter-payment queue through online payment, removing manual data entry through QR check-in, and running AI triage in parallel with, rather than after, vitals collection.

Scenario	Estimated Total Time	Estimated Time Saved
Visit without eye tests	23–27 minutes	~15–21 minutes
Visit with eye tests	33–42 minutes	~20–36 minutes

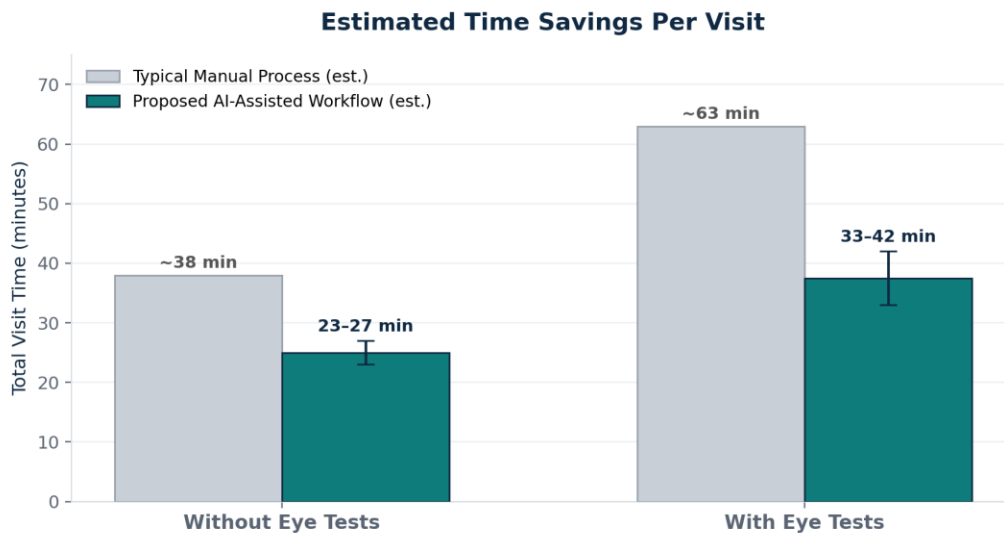


Figure 9.1 — Estimated total visit time: typical manual process vs. proposed AI-assisted workflow. The manual-process bars are an illustrative planning baseline, not measured Hashmanis data.

Where the Estimated Time Comes From

- Online payment: removes an estimated 5–8 minutes of counter and cashier queueing.
- QR check-in: removes manual record lookup and data entry, saving roughly 3–5 minutes.
- Parallel AI triage: avoids a separate triage queue, saving a further 5–10 minutes, with the larger figure applying to visits that also need eye tests.

Note: All figures in this section are planning estimates to be validated through a future pilot. They are not measured results from a live deployment.

10. Benefits

Shorter Waiting Times

AI-assisted sorting means urgent cases are seen first rather than waiting behind routine visits purely due to arrival order.

Less Administrative Work

Digital receipts, online payment, and QR check-in remove manual queues and repetitive data entry.

Better Patient Flow

Parallelizing triage with vitals collection avoids adding a separate review queue to the visit.

Faster Handling of Urgent Cases

Emergency and urgent labels route patients to the next available doctor ahead of routine cases.

Digital Check-in & Receipts

QR-based check-in and digital payment records reduce paperwork for both patients and staff.

11. Safety & Privacy

- Doctor remains in charge: every diagnosis and treatment decision is made by the attending doctor, not the AI module.
- AI only assists: the triage module produces a suggested label and routing, functioning as decision support rather than a decision-maker.
- Doctor can override: any AI triage label can be confirmed or overridden at the consultation stage, with the override taking precedence immediately.
- Secure patient records: all patient data is proposed to remain within existing secure hospital record systems.
- Authorized access only: access to patient data is restricted to authorized clinical and administrative staff.
- Patient data privacy: the triage module uses data already collected for clinical care and does not introduce new external data sharing.

12. System Architecture

The proposed architecture mirrors the patient workflow directly, so that each stage of the visit corresponds to one component in the system.

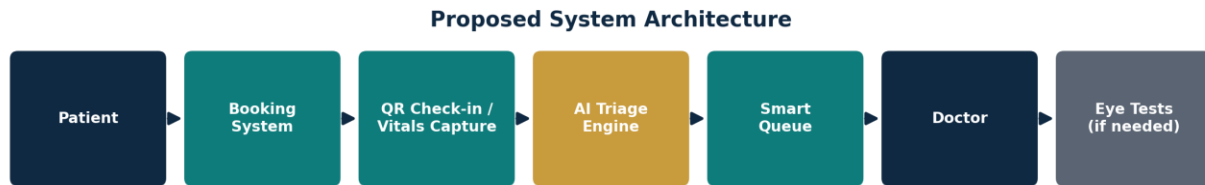


Figure 12.1 — Proposed system architecture: Patient → Booking → QR Check-in / Vitals → AI Triage → Smart Queue → Doctor → Eye Tests.

Patient data flows forward through booking, check-in, and triage; the doctor's decision, in turn, is the point at which the smart queue's suggested routing is confirmed or corrected before any further testing is ordered.

13. Functional Requirements

ID	Requirement
FR-1	The system shall allow patients to book an ophthalmology appointment and pay online.
FR-2	The system shall generate a QR code for each booking to enable check-in on arrival.
FR-3	The system shall capture recorded vitals into the patient's digital record at check-in.

ID	Requirement
FR-4	The system shall automatically classify each patient as emergency, urgent, or routine based on symptoms, vitals, and history.
FR-5	The system shall route patients into a smart queue according to their assigned priority label.
FR-6	The system shall present the doctor with the AI triage label and underlying data before consultation.
FR-7	The system shall allow the doctor to confirm or override the AI triage label, and shall log the outcome.
FR-8	The system shall record the doctor's diagnosis and treatment decision, including any ordered eye tests.
FR-9	The system shall generate a digital receipt for each completed payment.

14. Non-Functional Requirements

ID	Requirement
NFR-1	Availability — the AI triage module should respond in under one minute so it does not add a separate waiting step.
NFR-2	Reliability — a manual fallback process must remain available in case of system downtime.
NFR-3	Security — patient data must be encrypted in transit and at rest, with access limited to authorized staff.
NFR-4	Usability — the booking and check-in flow should be usable by patients with limited digital literacy, with staff-assisted alternatives available.
NFR-5	Scalability — the design should support deployment across multiple Hashmanis branches without redesign.
NFR-6	Auditability — every AI triage label and every doctor override must be logged for later review.
NFR-7	Maintainability — triage rules should be updatable by authorized staff as clinical needs evolve.

15. AI Decision Flow

The figure below traces how the AI triage module moves from raw inputs to a classification and, ultimately, to a doctor-confirmed outcome.

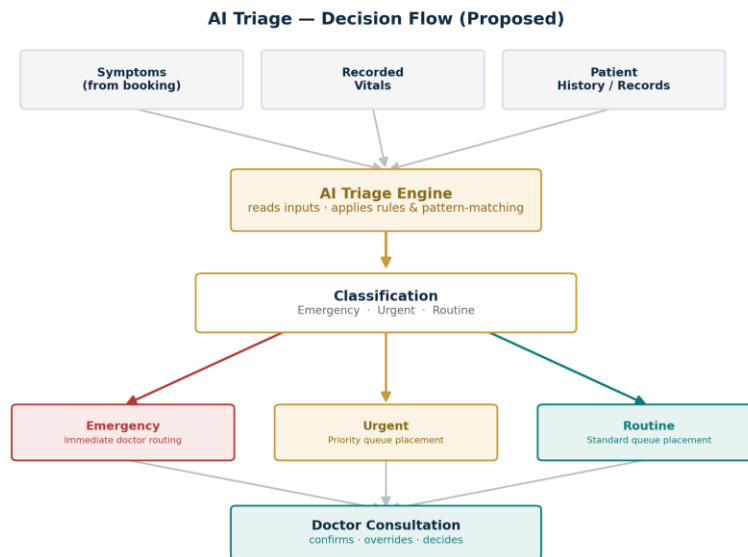


Figure 15.1 — Proposed AI triage decision flow, from inputs through classification to mandatory doctor review.

Every path through this flow — regardless of the assigned label — terminates at doctor consultation. There is no route by which a patient is treated, tested, or discharged purely on the basis of the AI classification.

16. Data Flow

The data flow diagram below shows how information moves between the booking, vitals, and history sources, the secure record store, the AI triage module, and the doctor.

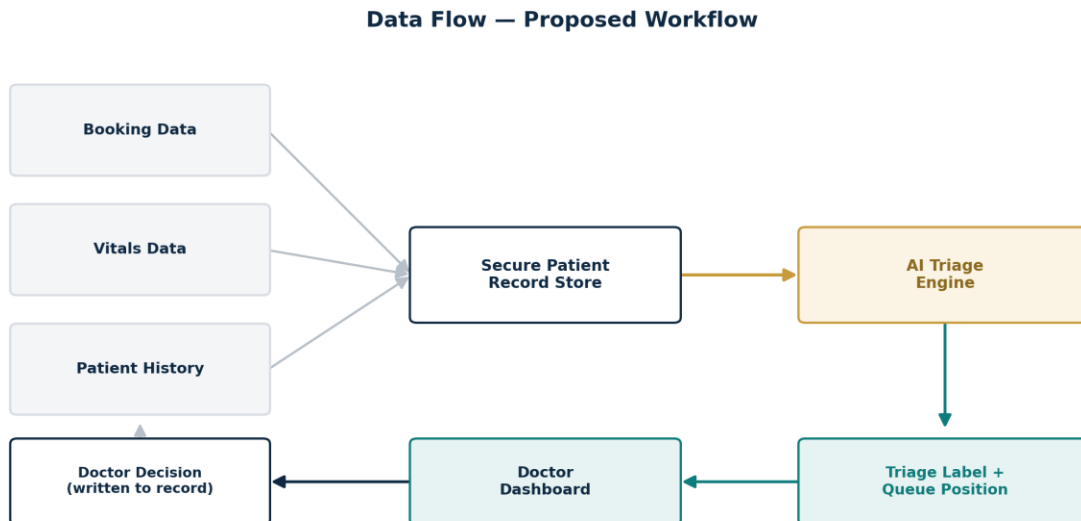


Figure 16.1 — Proposed data flow: patient-originated data feeds a secure record store, which supplies the AI triage module; the doctor's decision is written back to the same record.

17. Risk, Safety & Limitations

Limitations

This proposal is a system design, not a validated clinical tool. Its triage logic has not been tested against real patient outcomes, its timing figures are planning estimates rather than measured data, and its accuracy for rare or atypical presentations is currently unknown. These limitations are the reason a bounded pilot, described in Section 19, is recommended before any wider rollout.

Risks & Mitigations

Risk	Mitigation
AI mis-classifies a genuinely urgent case as routine	Doctor review remains mandatory for every patient regardless of AI label; staff treat the label as a suggestion, and worsening symptoms during waiting can trigger immediate re-review.
Patients without smartphones or digital literacy struggle with online booking or QR check-in	Staffed check-in counters remain available at every branch as a parallel path; the digital flow is optional, not required to be seen.
System downtime disrupts check-in or triage	A manual fallback process stays fully operational, and staff are trained to switch to it without disrupting patient flow.
Patient data privacy or security concerns	Triage uses data already collected for clinical care, within existing secure systems, with authorized access only.
Triage rules do not generalize well to rare or atypical presentations	Rules are reviewed periodically against override patterns, and doctors are explicitly instructed not to rely on the label alone for atypical cases.

18. Expected Outcomes

- An estimated 15–21 minutes saved per visit for patients who do not require eye tests.
- An estimated 20–36 minutes saved per visit for patients who do require eye tests.
- Faster identification and routing of emergency and urgent ophthalmic cases.
- Reduced administrative burden on front-desk and clinical staff through digital booking, payment, and check-in.
- A consistent triage approach that can, if validated, be applied across multiple Hashmanis branches.

Note: These outcomes are projected, planning-stage estimates. They are intended to be validated — and revised if necessary — through the pilot described in Section 19.

19. Future Enhancements

- Pilot the workflow at a single Hashmanis branch for ophthalmology only, with the manual process kept fully available as a fallback.
- Track actual step-by-step timings against the estimates in Section 9 to validate or revise projected savings.
- Log every doctor override of the AI triage label to build an evidence base on where the model is, and isn't, reliable.
- Refine triage rules using override patterns and clinical staff feedback before considering a wider rollout.
- Extend the validated design to other high-urgency-spread specialties beyond ophthalmology, if the pilot outcomes support it.
- Explore patient-facing status updates (e.g., estimated wait time) once queue routing has been validated in practice.

20. Conclusion

This document has proposed an AI-assisted patient workflow for Hashmanis Hospital's ophthalmology service line, built around seven stages that separate automatable administrative and sorting work from clinical judgement. Online booking, QR check-in, and AI-assisted triage are proposed to reduce total visit time by an estimated 15 to 36 minutes, while every diagnostic and treatment decision remains firmly with the attending doctor.

The design is deliberately conservative in scope: it is limited to a single specialty, keeps AI to one clearly bounded decision-support role, and treats every timing figure as an estimate to be tested rather than a guarantee. The recommended next step is a bounded, single-branch pilot that validates these assumptions against real patient flow before any broader deployment is considered.